

GCSE Mathematics

Predicted Paper 3 — Higher Tier

Set D

Pearson Edexcel style (1MA1 / 3H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. Estimate the value of $(38.4 \times 6.97) / 0.193$ by rounding each number to 1 significant figure. Show your working.

(3 marks)

2. Without using a calculator, work out $2 \frac{1}{3} - \frac{4}{5}$. Give your answer as a mixed number in its simplest form.

(3 marks)

3. The first four terms of an arithmetic sequence are 7, 12, 17, 22, ...

(a) Find an expression, in terms of n , for the n th term of this sequence.

(2 marks)

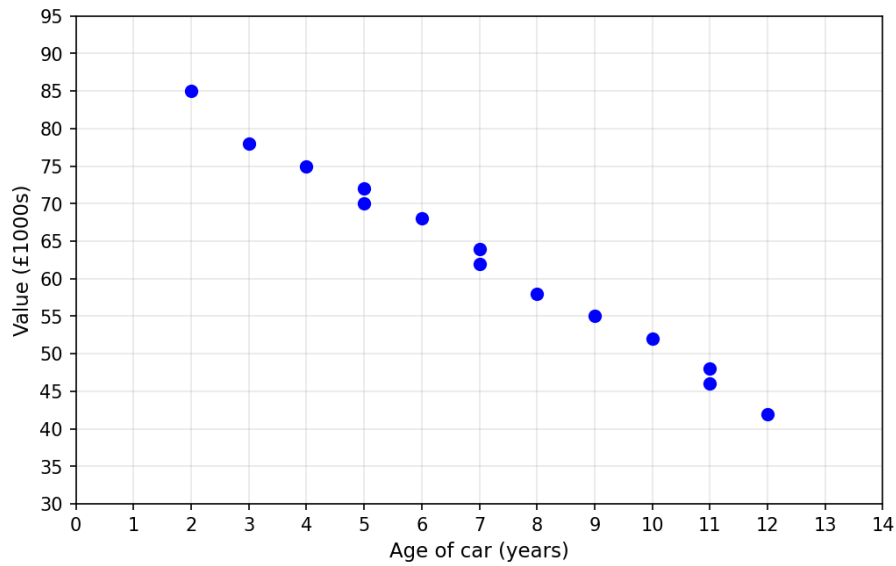
(b) Is 200 a term in this sequence? You must show your working.

(2 marks)

4. A map has a scale of 1 : 25,000. Two villages are 12 cm apart on the map. Calculate the actual distance between the two villages in kilometres.

(3 marks)

5. The scatter graph shows the age and value of 14 used cars.



(a) Describe the relationship between the age of a car and its value.

(1 marks)

(b) A car is 6 years old. Use the scatter graph to estimate its value.

(2 marks)

(c) Explain why your estimate in part (b) may not be reliable for cars older than 12 years.

(2 marks)

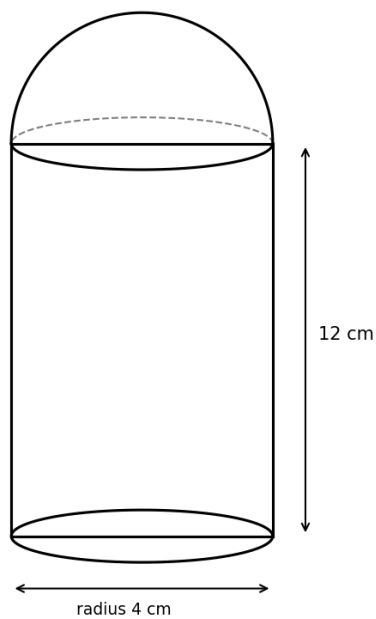
6. A shopkeeper buys 200 items for £1,400 in total. He sells 180 of them for £12 each. The remaining 20 items are sold off at half the selling price. Calculate the shopkeeper's overall percentage profit.

(4 marks)

7. Solve $3x^2 - 5x - 1 = 0$. Give your answers correct to 3 significant figures.

(3 marks)

8. The diagram shows a solid shape made from a cylinder with a hemisphere on top. The cylinder has radius 4 cm and height 12 cm. The hemisphere has radius 4 cm.



(a) Work out the total volume of the shape. Give your answer in terms of π .

(3 marks)

(b) Work out the total surface area of the shape (curved surface of cylinder + curved surface of hemisphere + bottom of cylinder). Give your answer to 3 significant figures.

(4 marks)

9. A school has 850 students: 400 in Years 7-9, 300 in Years 10-11, and 150 in Years 12-13. The head teacher wants to survey a stratified sample of 85 students.

(a) Work out the number of students that should be sampled from each year group.

(3 marks)

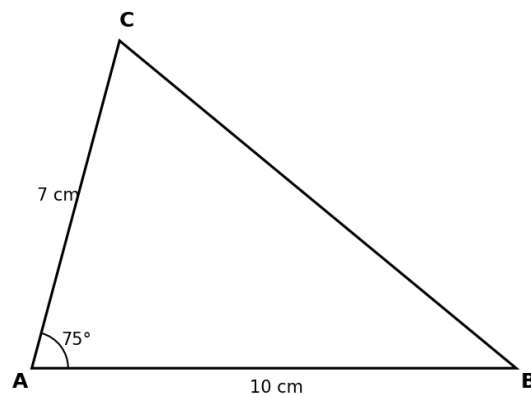
(b) Explain one advantage of using a stratified sample rather than a simple random sample.

(1 marks)

10. Expand and simplify $(3 + \sqrt{5})(2 - \sqrt{5})$. Give your answer in the form $a + b\sqrt{5}$ where a and b are integers.

(3 marks)

11. In triangle ABC, $AB = 10$ cm, $AC = 7$ cm and angle $BAC = 75^\circ$.



(a) Calculate the length of BC. Give your answer to 3 significant figures.

(3 marks)

(b) Calculate the area of triangle ABC. Give your answer to 3 significant figures.

(2 marks)

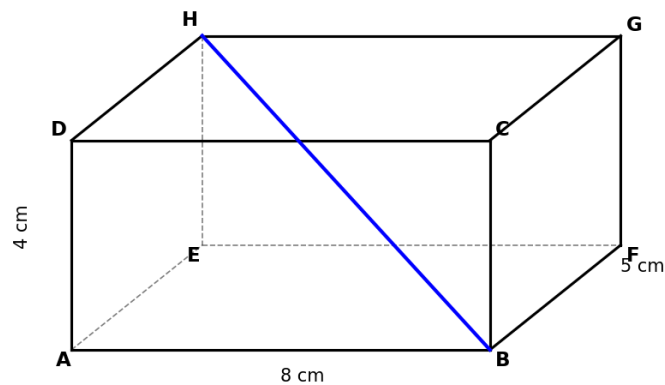
12. Show that the recurring decimal 0.318181818... (with the digits 18 repeating after 0.3) can be written as the fraction $\frac{7}{22}$.

(4 marks)

13. Solve $(x + 3) / 4 = (2x - 1) / 3$.

(3 marks)

14. The diagram shows cuboid ABCDEFGH with dimensions AB = 8 cm, BC = 5 cm, and CG = 4 cm.



(a) Calculate the length of the diagonal BH of the cuboid. Give your answer correct to 3 significant figures.

(3 marks)

(b) Calculate the size of the angle between the diagonal BH and the base ABCD. Give your answer correct to 1 decimal place.

(3 marks)

15. $f(x) = 3x - 1$ and $g(x) = (x + 2)/4$.

(a) Find $fg(x)$. Simplify your answer.

(3 marks)

(b) Show that $f^{-1}(x) = g^{-1}(x)$ has solution $x = 7/11$.

(4 marks)

16. A rectangle has length 12.5 cm and width 7.4 cm, each measured to the nearest 0.1 cm.

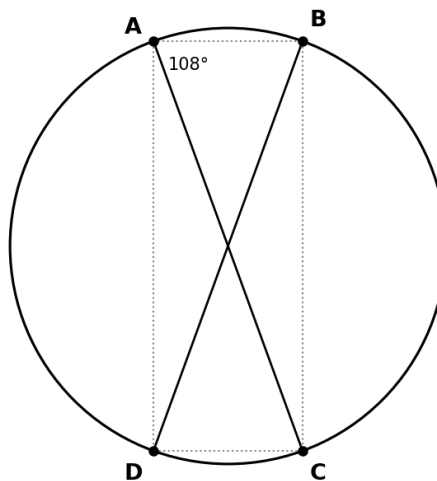
(a) Write down the upper bound for the length.

(1 marks)

(b) Calculate the lower bound and upper bound for the area of the rectangle.

(3 marks)

17. ABCD is a quadrilateral with all four vertices on the circumference of a circle. Angle DAB = 108° .



(a) Work out the size of angle DCB. Give a reason for your answer.

(2 marks)

(b) Explain why ABCD cannot have two right angles at A and C.

(2 marks)

18. Liam drives the first 60 km of a journey at an average speed of 80 km/h. He drives the remaining 40 km at an average speed of 50 km/h. Calculate his average speed for the entire 100 km journey. Give your answer to 1 decimal place.

(4 marks)

19. Events X and Y are NOT mutually exclusive. $P(X) = 0.4$, $P(Y) = 0.3$, and $P(X \cap Y) = 0.1$.

(a) Find $P(X \cup Y)$.

(2 marks)

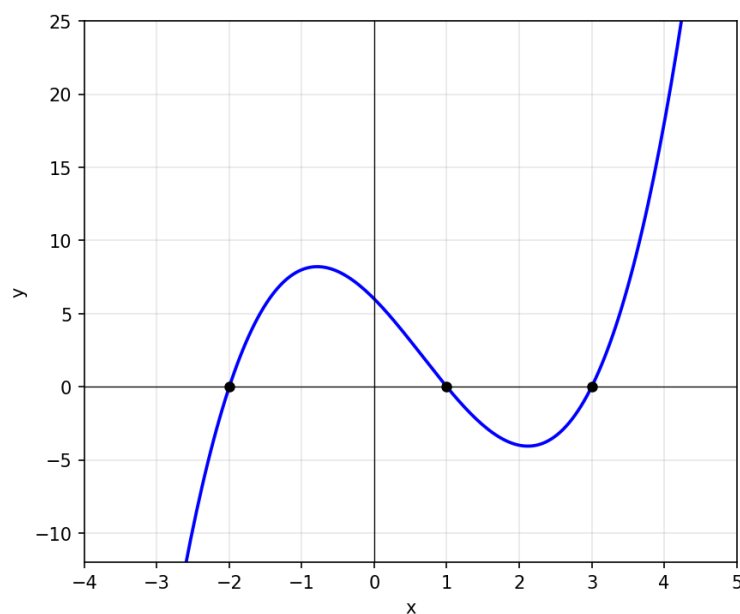
(b) Find $P(\text{neither } X \text{ nor } Y)$.

(2 marks)

(c) Are X and Y independent events? Justify your answer with calculation.

(2 marks)

20. The graph shows the curve $y = (x - 1)(x + 2)(x - 3)$.



(a) Write down the coordinates of the three points where the curve crosses the x-axis.

(2 marks)

(b) Write down the y-intercept of the curve.

(1 marks)

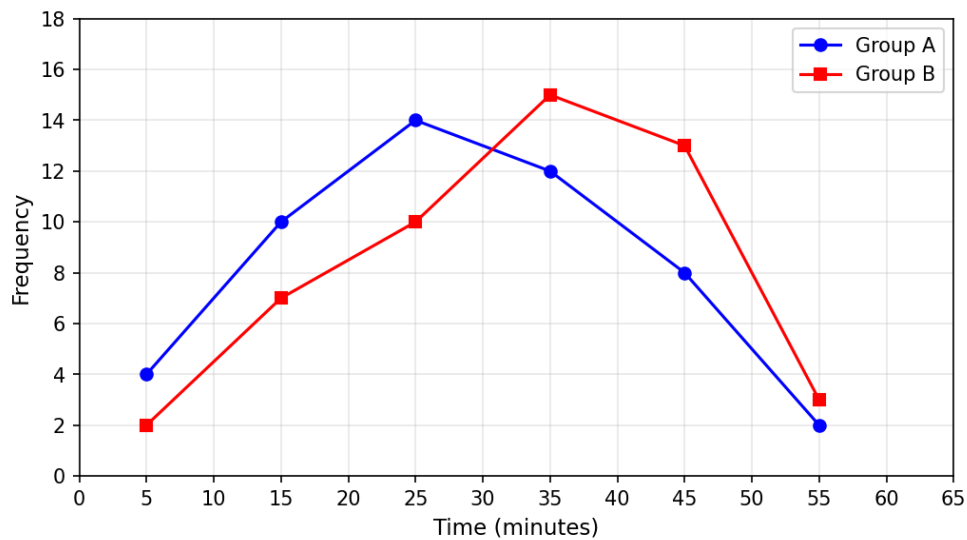
(c) For what values of x is $y > 0$? Give your answer using inequalities.

(2 marks)

21. Solve $(2x - 7) / 3 + x = 5$.

(3 marks)

22. The frequency polygons show the times (in minutes) taken by Group A and Group B to complete a task.



(a) Calculate an estimate for the mean time taken by Group A.

(3 marks)

(b) Without further calculation, state with reason which group has a higher mean time.

(2 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 3 Set D

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	≈ 1400	M1: rounds each: 40, 7, 0.2. M1: $(40 \times 7)/0.2 = 280/0.2$. A1: 1400. [3]
2	$1 \frac{8}{15}$	M1: common denominator 15: $35/15 - 12/15$. M1: $= 23/15$. A1: $1 \frac{8}{15}$. [3]
3(a)	nth term = $5n + 2$	M1: identifies common difference = 5. A1: $5n + 2$ (or equivalent). [2]
3(b)	No, 200 is not a term	M1: $5n + 2 = 200 \Rightarrow 5n = 198$. A1: $n = 39.6$ (not an integer), so 200 is not in the sequence. [2]
4	3 km	M1: $12 \times 25000 = 300,000$ cm. M1: converts to km: $300,000$ cm = 3000 m = 3 km. A1: 3 km. [3]
5(a)	Negative correlation	B1: states negative correlation (as age increases, value decreases). [1]
5(b)	Around £68,000 (accept 65,000–72,000)	M1: reads from graph at $x = 6$. A1: value of y between £65k and £72k. [2]
5(c)	Extrapolation beyond the data range	M1: identifies that 12 years is the highest x -value in the data. A1: states extrapolating beyond the data range is unreliable as the trend may not continue. [2]
6	62.9% (1 d.p.)	M1: revenue: 180 sold at £12 = £2,160; 20 sold at £6 = £120. M1: total revenue = £2,280. M1: profit = $2280 - 1400 = £880$. A1: $880/1400 \times 100 = 62.9\%$ (1 d.p.). [4]
7	$x = 1.85$ or $x = -0.180$ (3 s.f.)	M1: $a = 3$, $b = -5$, $c = -1$. M1: discriminant = $25 + 12 = 37$. A1: $x = (5 \pm \sqrt{37})/6 \Rightarrow x \approx 1.85$ or $x \approx -0.18$. [3]
8(a)	$(704\pi)/3 \text{ cm}^3$	M1: cylinder volume = $\pi r^2 h = \pi(4)^2(12) = 192\pi$. M1: hemisphere volume = $(1/2)(4/3)\pi r^3 = (2/3)\pi(4)^3 = 128\pi/3$. A1: total = $192\pi + 128\pi/3 = (576\pi + 128\pi)/3 = 704\pi/3$. [3]
8(b)	452 cm^2 (3 s.f.)	M1: curved cylinder = $2\pi r h = 2\pi(4)(12) = 96\pi$. M1: curved hemisphere = $(1/2)(4\pi r^2) = 2\pi(16) = 32\pi$. M1: bottom circle = $\pi r^2 = 16\pi$. A1: total = $144\pi = 452.4 \approx 452 \text{ cm}^2$ (3 s.f.). [4]
9(a)	Y7-9: 40; Y10-11: 30; Y12-13: 15	M1: $85/850 = 0.1$ (sampling fraction). M1: applies to each year group. A1: 40, 30, 15. [3]
9(b)	Representative across year groups	B1: stratified sampling ensures each year group is represented in proportion to its size in the population. [1]
10	$1 - \sqrt{5}$	M1: expands: $6 - 3\sqrt{5} + 2\sqrt{5} - 5$. M1: simplifies: $1 - \sqrt{5}$. A1: $a = 1$, $b = -1$. [3]
11(a)	$BC \approx 10.5 \text{ cm}$ (3 s.f.)	M1: cosine rule: $BC^2 = 10^2 + 7^2 - 2(10)(7)\cos 75^\circ$. M1: $= 100 + 49 - 140(0.2588) = 112.77$. A1: $BC = \sqrt{112.77} = 10.6 \text{ cm}$ (3 s.f.). [3]
11(b)	Area $\approx 33.8 \text{ cm}^2$ (3 s.f.)	M1: area = $(1/2)(10)(7)\sin 75^\circ$. A1: $= 35 \times 0.9659 \approx 33.8 \text{ cm}^2$. [2]
12	Shown: $0.3181818... = 7/22$	M1: lets $x = 0.3181818...$, so $10x = 3.181818...$ and $1000x = 318.1818...$. M1: $1000x - 10x = 315 \Rightarrow 990x = 315$. M1: $x = 315/990$. A1: simplifies $315/990 = 63/198 = 7/22$. ✓ [4]
13	$x = 13/5$	M1: cross-multiplies: $3(x + 3) = 4(2x - 1)$. M1: $3x + 9 = 8x - 4 \Rightarrow 13 = 5x$. A1: $x = 13/5$ (or 2.6). [3]
14(a)	$BH \approx 10.2 \text{ cm}$ (3 s.f.)	M1: $BD^2 = 8^2 + 5^2 = 89$ (face diagonal of base). M1: $BH^2 = BD^2 + 4^2 = 89 + 16 = 105$. A1: $BH = \sqrt{105} \approx 10.2 \text{ cm}$. [3]
14(b)	22.9° (1 d.p.)	M1: angle of interest is at B in right triangle BDH ($DH = CG = 4$, $BD = \sqrt{89}$). M1: $\tan(\text{angle DBH}) = 4/\sqrt{89} = 0.4241$. A1: angle = $\tan^{-1}(0.4241) = 23.0^\circ$ (1 d.p.). Accept 22.9° to 23.0° depending on precision. [3]

Q	Answer	Mark scheme
15(a)	$fg(x) = (3x + 2)/4$	M1: $f((x + 2)/4) = 3((x + 2)/4) - 1$. M1: $= (3(x + 2))/4 - 1 = (3x + 6 - 4)/4$. A1: $fg(x) = (3x + 2)/4$. [3]
15(b)	$x = 7/11$	M1: $f^{-1}(x) = (x + 1)/3$. M1: $g^{-1}(x) = 4x - 2$. M1: sets equal: $(x + 1)/3 = 4x - 2 \Rightarrow x + 1 = 12x - 6$. A1: $7 = 11x \Rightarrow x = 7/11$. [4]
16(a)	12.55 cm	B1: upper bound = $12.5 + 0.05$. [1]
16(b)	Lower: 91.6275 cm^2 ; Upper: 93.0225 cm^2	M1: lower bounds: 12.45×7.35 . M1: upper bounds: 12.55×7.45 . A1: states both lower 91.51–91.63 and upper 92.95–93.02. [3]
17(a)	Angle DCB = 72°	B1: 72° . B1: reason "opposite angles of cyclic quadrilateral sum to 180° " ($180 - 108 = 72$). [2]
17(b)	Two right angles would sum to 180°	M1: identifies that opposite angles must sum to 180. A1: states that if A and C are opposite and both equal 90° , they sum to 180° , but the problem says angle A = 108° , so C must be 72° . Alternatively: if A = C = 90° , then D and B must also sum to 180° , which is consistent, but only if A = 90° specifically. [2]
18	64.5 km/h (1 d.p.)	M1: time for first leg = $60/80 = 0.75$ h. M1: time for second leg = $40/50 = 0.8$ h. M1: total time = $0.75 + 0.8 = 1.55$ h. A1: average speed = $100/1.55 = 64.5$ km/h (1 d.p.). [4]
19(a)	$P(X \cup Y) = 0.6$	M1: addition rule $P(X) + P(Y) - P(X \cap Y) = 0.4 + 0.3 - 0.1$. A1: 0.6. [2]
19(b)	0.4	M1: $1 - P(X \cup Y) = 1 - 0.6$. A1: 0.4. [2]
19(c)	Not independent	M1: tests $P(X) \times P(Y) = 0.4 \times 0.3 = 0.12$. A1: $0.12 \neq P(X \cap Y) = 0.1$, so events are NOT independent. [2]
20(a)	$(-2, 0)$, $(1, 0)$, $(3, 0)$	B1: identifies roots from factored form. B1: full set of coordinates. [2]
20(b)	y-intercept: $(0, 6)$	B1: substitutes $x = 0$: $y = (-1)(2)(-3) = 6$. [1]
20(c)	$-2 < x < 1$ or $x > 3$	M1: identifies intervals between/beyond roots where curve is above x-axis. A1: full solution $-2 < x < 1$ or $x > 3$. [2]
21	$x = 22/5$	M1: multiplies through by 3: $(2x - 7) + 3x = 15$. M1: $5x - 7 = 15 \Rightarrow 5x = 22$. A1: $x = 22/5$ (or 4.4). [3]
22(a)	28.2 minutes	M1: midpoints $\times$ frequencies: $5(4) + 15(10) + 25(14) + 35(12) + 45(8) + 55(2)$. M1: $\Sigma fx = 20 + 150 + 350 + 420 + 360 + 110 = 1410$; $\Sigma f = 50$. A1: $1410/50 = 28.2$ min. [3]
22(b)	Group B has a higher mean	M1: identifies that Group B's frequencies are weighted toward higher times (peak at 35 vs Group A's peak at 25). A1: explains why this implies a higher mean. [2]

Worked Solutions — Predicted Paper 3 Set D

Higher Tier · Edexcel 1MA1 style

Q1: Estimation

Round each to 1 s.f.: $38.4 \rightarrow 40$; $6.97 \rightarrow 7$; $0.193 \rightarrow 0.2$.

Estimate = $(40 \times 7) / 0.2 = 280 / 0.2 = \mathbf{1400}$.

Q2: Fractions

Common denominator 15: $2 \frac{1}{3} = \frac{7}{3} = \frac{35}{15}$; $\frac{4}{5} = \frac{12}{15}$.

$\frac{35}{15} - \frac{12}{15} = \frac{23}{15}$.

Convert to mixed number: $23 \div 15 = 1$ remainder 8 = $\mathbf{1 \frac{8}{15}}$.

Q3: Arithmetic sequence

(a) First term = 7, common difference = 5. nth term = $7 + (n - 1)(5) = \mathbf{5n + 2}$.

(b) $5n + 2 = 200 \Rightarrow 5n = 198 \Rightarrow n = 39.6$.

Since n must be a positive integer, **200 is not a term** in this sequence.

Q4: Map scale

12 cm on map $\times 25,000 = 300,000$ cm in reality.

Convert to km: $300,000$ cm = $3,000$ m = $\mathbf{3 \text{ km}}$.

Q5: Scatter graph

(a) **Negative correlation**: as age increases, value decreases.

(b) Reading at age = 6: value $\approx \mathbf{\pounds 68,000}$ (accept $\pounds 65\text{k} - \pounds 72\text{k}$).

(c) 12 years is the maximum age in the data set. Using the trend beyond 12 years is **extrapolation**, which is unreliable because the relationship may change or break down beyond the observed range.

Q6: Mixed-price profit

180 items sold at full price $\pounds 12$: revenue = $180 \times 12 = \pounds 2,160$.

20 items sold at half price = $\pounds 6$ each: revenue = $20 \times 6 = \pounds 120$.

Total revenue = $2,160 + 120 = \pounds 2,280$.

Profit = $2,280 - 1,400 = \pounds 880$.

Percentage profit = $(880 / 1,400) \times 100 \approx \mathbf{62.9\%}$ (1 d.p.).

Q7: Quadratic formula

$a = 3$, $b = -5$, $c = -1$.

Discriminant = $(-5)^2 - 4(3)(-1) = 25 + 12 = 37$.

$x = (5 \pm \sqrt{37}) / 6 = (5 \pm 6.083) / 6$.

Positive root: $x = 11.083 / 6 = \mathbf{1.85}$ (3 s.f.).

Negative root: $x = -1.083 / 6 = \mathbf{-0.180}$ (3 s.f.).

Q8: Composite solid

(a) Cylinder volume = $\pi r^2 h = \pi(4)^2(12) = 192\pi \text{ cm}^3$.

Hemisphere volume = $(1/2)(4/3)\pi r^3 = (2/3)\pi(4)^3 = (2/3)(64)\pi = 128\pi/3 \text{ cm}^3$.

Total = $192\pi + 128\pi/3 = (576\pi + 128\pi)/3 = (704\pi)/3 \text{ cm}^3 (\approx 737 \text{ cm}^3)$.

(b) Curved cylinder = $2\pi rh = 2\pi(4)(12) = 96\pi$.

Curved hemisphere = $(1/2)(4\pi r^2) = 2\pi(16) = 32\pi$.

Bottom circle = $\pi r^2 = 16\pi$.

Total = $96\pi + 32\pi + 16\pi = 144\pi \approx 452 \text{ cm}^2$ (3 s.f.).

Q9: Stratified sampling

(a) Sampling fraction = $85 / 850 = 0.1$ (10%).

Y7-9: $10\% \times 400 = 40$; Y10-11: $10\% \times 300 = 30$; Y12-13: $10\% \times 150 = 15$.

Check: $40 + 30 + 15 = 85$. ✓

(b) Stratified sampling ensures each year group is represented in proportion to its size, so the sample better reflects the school as a whole.

Q10: Surds expansion

$(3 + \sqrt{5})(2 - \sqrt{5}) = 6 - 3\sqrt{5} + 2\sqrt{5} - (\sqrt{5})^2$.

$= 6 - \sqrt{5} - 5 = 1 - \sqrt{5}$.

So $a = 1$, $b = -1$.

Q11: Cosine rule

(a) $BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos(BAC)$.

$= 10^2 + 7^2 - 2(10)(7)\cos 75^\circ$

$= 100 + 49 - 140(0.2588) = 149 - 36.24 = 112.76$.

$BC = \sqrt{112.76} \approx 10.6 \text{ cm}$ (3 s.f.).

(b) Area = $(1/2)(AB)(AC)\sin(BAC) = (1/2)(10)(7)\sin 75^\circ = 35 \times 0.9659 \approx 33.8 \text{ cm}^2$ (3 s.f.).

Q12: Recurring decimal

Let $x = 0.3181818\ldots$

Then $10x = 3.181818\ldots$ (one digit moved before the repeat).

And $1000x = 318.181818\ldots$

Subtract: $1000x - 10x = 318.181818\ldots - 3.181818\ldots = 315$.

$990x = 315 \Rightarrow x = 315/990$.

Simplify: $\text{gcd}(315, 990) = 45$. So $315/990 = 7/22$.

✓ $x = 7/22$.

Q13: Equation with fractions

Cross-multiply: $3(x + 3) = 4(2x - 1)$.

$3x + 9 = 8x - 4 \Rightarrow 13 = 5x \Rightarrow x = 13/5$ (or 2.6).

Q14: 3D Pythagoras + angle

(a) Face diagonal $BD^2 = AB^2 + AD^2 = 8^2 + 5^2 = 64 + 25 = 89$.

Body diagonal $BH^2 = BD^2 + DH^2 = 89 + 4^2 = 105$.

$BH = \sqrt{105} \approx \mathbf{10.2 \text{ cm (3 s.f.)}}$.

(b) The angle between BH and the base is at B in right triangle BDH (with right angle at D).

$\tan(\text{angle DBH}) = DH / BD = 4 / \sqrt{89} = 0.4241$.

Angle = $\tan^{-1}(0.4241) = \mathbf{23.0^\circ (1 \text{ d.p.})}$.

Q15: Function composition

(a) $fg(x) = f(g(x)) = f((x+2)/4) = 3 \cdot (x+2)/4 - 1 = (3(x+2))/4 - 1$.

$= (3x + 6 - 4)/4 = \mathbf{(3x + 2)/4}$.

(b) Find inverses:

$f^{-1}(x) = (x+1)/3$ (from $y = 3x - 1$).

$g^{-1}(x) = 4x - 2$ (from $y = (x+2)/4$).

Set equal: $(x+1)/3 = 4x - 2$.

$x + 1 = 12x - 6 \Rightarrow 7 = 11x \Rightarrow \mathbf{x = 7/11. \checkmark}$

Q16: Bounds

(a) Length upper bound = $12.5 + 0.05 = \mathbf{12.55 \text{ cm}}$.

(b) Lower bound area = $12.45 \times 7.35 = \mathbf{91.5075 \text{ cm}^2}$.

Upper bound area = $12.55 \times 7.45 = \mathbf{93.5 \text{ cm}^2}$.

(Exact: lower = 91.5075, upper = 93.4975.)

Q17: Cyclic quadrilateral

(a) Opposite angles of a cyclic quadrilateral sum to 180° .

Angle DCB = $180 - \text{angle DAB} = 180 - 108 = \mathbf{72^\circ}$.

(b) Opposite angles in a cyclic quadrilateral sum to 180° . If A and C are both 90° , they sum to 180° (consistent with the cyclic property). However, in this specific case angle A = 108° , so angle C must be 72° . The two right angles at A and C are **not possible while keeping angle A = 108°** .

Q18: Average speed

Time for first part = distance/speed = $60/80 = 0.75$ hours.

Time for second part = $40/50 = 0.8$ hours.

Total distance = 100 km. Total time = $0.75 + 0.8 = 1.55$ hours.

Average speed = $100 / 1.55 \approx \mathbf{64.5 \text{ km/h (1 d.p.)}}$.

Q19: Probability of combined events

(a) $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y) = 0.4 + 0.3 - 0.1 = \mathbf{0.6}$.

(b) $P(\text{neither}) = 1 - P(X \cup Y) = 1 - 0.6 = \mathbf{0.4}$.

(c) For independence: $P(X) \times P(Y) = 0.4 \times 0.3 = 0.12$.

But $P(X \cap Y) = 0.1$, and $0.12 \neq 0.1$.

So X and Y are **not independent**.

Q20: Cubic graph

(a) Roots from factored form: $x = 1$, $x = -2$, $x = 3$.

Coordinates: **$(-2, 0)$, $(1, 0)$, $(3, 0)$** .

(b) y-intercept: substitute $x = 0$: $y = (-1)(2)(-3) = 6$. So $(0, 6)$.

(c) From the graph, $y > 0$ for x between the leftmost root and the middle root, and to the right of the rightmost root:

$-2 < x < 1$ or $x > 3$.

Q21: Linear equation

$(2x - 7)/3 + x = 5$.

Multiply through by 3: $(2x - 7) + 3x = 15$.

$5x - 7 = 15 \Rightarrow 5x = 22 \Rightarrow x = 22/5$ (or 4.4).

Q22: Frequency polygon

(a) Midpoints $\times$ frequencies (Group A): $5(4) + 15(10) + 25(14) + 35(12) + 45(8) + 55(2)$.

$= 20 + 150 + 350 + 420 + 360 + 110 = 1410$.

$\Sigma f = 4 + 10 + 14 + 12 + 8 + 2 = 50$.

Mean for Group A = $1410 / 50 = 28.2$ minutes.

(b) Group B's peak frequency is at the 35-minute midpoint, whereas Group A's peak is at the 25-minute midpoint. Group B's distribution is shifted toward higher times overall, so **Group B has a higher mean time**.